Topic 5 - Arms Race

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Introduction

This section should outline the real-world background and context of your model. Discuss the motivation one might have to study this model. You may want to research the relevant literature and include some citations to support your framing of the real-world context (depending on the project). Be sure to include only background information and context that is directly related to your research project. Really focus on what the reader needs to know to understand your work, and be interested and excited by it.

The final paragraph of your Introduction should frame your research question and de- scribe what you will be studying in your paper. Briefly mention, in words, what you will be doing and showing in the rest of the paper. It is often good to lay out your main find- ing/result here, too. A good Introduction section for an applied mathematics paper ideally has no mathematical symbols or jargon; instead, it should be focused on the real-world con- text and what you are trying to study. The math is a tool you will introduce later, not the main focus of the paper.

Consider @Talele2025, a great guy who wrote some cool stuff.

Base Model

For either hypothetical country, P and G, the assumptions for our base model are as follows:

1. The rate of change in one nation's arms spending is directly proportional to the other nation's arms spending
2. The rate of change in one nation's arms spending is also negatively directly proportion to that nation's own current spending
3. There may be underlying trust and hostility between the two nations, which will have a constant effect on rate of change
4. For now, we assume that each country has infinite amounts of wealth at their disposal (not realistic at all in a real-world context)

Based on these assumptions, the following model was derived:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r \tag{1}$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s \tag{2}$$

In particular $\frac{\partial p}{\partial t}, \frac{\partial g}{\partial t}$ denote the rate of change of each country's arm spending. The variables a, c measure the effect of mutual fear that each country has on the other's arms spending, b, d measure the effect of expenditure burden that each country has on their own spending, and r, s measure the underlying hostility or trust that the each nation has to another.

For the sake of realism, $p(t), g(t)$ are held to be non-negative values as negative arms spending would not make sense in this scenario. Moreover, to ensure that mutual fear and expenditure burden match the behavior as laid out in the assumptions, a, b, c, d are held to be non-negative as well. On the other hand, r, s can assume any value where negative values denote trust and positive values denote hostility.

The parameters in the model (a, b, c, d, r, s) assume no form of units as they serve as abstractions of qualitative measures (i.e. it would be hard to attach a unit to expenditure burden or fear). However, the state variables themselves would be expressed in some common currency (e.g. U.S. dollars) and the derivatives of those state variables $(\frac{\partial p}{\partial t}, \frac{\partial g}{\partial t})$ would be expressed in that currency over time (e.g. U.S. dollars per year).

Theoretically speaking, this model is a system of multivariate, autonomous linear differential equations. In this situation, the Jacobian would have to be computed for the purposes of stability analysis.

Analysis of Base Model

To facilitate the stability analysis of the base model, we now reformulate the system of equations into matrix form:

$$\begin{bmatrix} \frac{\partial p}{\partial t} \\ \frac{\partial g}{\partial t} \end{bmatrix} = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix} \begin{bmatrix} p(t) \\ g(t) \end{bmatrix} + \begin{bmatrix} r \\ s \end{bmatrix}$$

Analysis without underlying trust or hostility

As a starting point, we assume that both nations have no underlying trust or hostility such that $r = s = 0$. Thus, the model equations are now:

$$\begin{bmatrix} \frac{\partial p}{\partial t} \\ \frac{\partial g}{\partial t} \end{bmatrix} = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix} \begin{bmatrix} p(t) \\ g(t) \end{bmatrix}$$

Note that by having the system in linear form, the Jacobian is simply the matrix

$$J = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix}$$

Setting $r = s = 0$ lends a unique advantage in that the resulting system of linear differential equations is now homogenous (i.e. there is no constant term). This means that a closed-form of the state-variables can be found in the form of:

$$\begin{bmatrix} p(t) \\ g(t) \end{bmatrix} = P \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} P^{-1} \begin{bmatrix} p(0) \\ g(0) \end{bmatrix}$$

Where λ_1, λ_2 denotes the eigenvalues of the Jacobian J, P denotes the matrix of eigenvectors, and $p(0), g(0)$ denotes the initial spending of the two countries.

To determine equilibrium points, set

$$\begin{bmatrix} \frac{\partial p}{\partial t} \\ \frac{\partial g}{\partial t} \end{bmatrix} = 0$$

Thus,

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix} \begin{bmatrix} p(t) \\ g(t) \end{bmatrix}$$

An obvious equlibrium that can be derived from this formulation is:

$$\begin{bmatrix} p^*(t) \\ g^*(t) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This equilibrium point corresponds to a scenario where both nations mutually demilitarize.

To determine the stability conditions of such a scenario, we need to compute the Jacobian. Since the system of differential equations are already in linear form, the Jacobian is simply given by:

$$J = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix}$$

In order for the scenario of mutual disarmament to be asymptotically stable, the eigenvalues of the jacobian should both be negative. The eigenvalues are given by:

$$\det(J - I\lambda) = \begin{bmatrix} -b-\lambda & a \\ c & -d-\lambda \end{bmatrix} = \lambda^2 + (b+d)\lambda + bd - ac = 0$$

The solutions to these eigenvalues are of the form:

$$\lambda = \frac{-(b+d) \pm \sqrt{(b+d)^2 - 4(bd-ac)}}{2}$$

Expanding the square term in the discriminant and simplifying the equation yields the following:

$$\lambda = \frac{-(b+d) \pm \sqrt{(b-d)^2 + 4ac}}{2}$$

It should be noted that because of the non-negative restrictions imposed upon a, b, c, d, the discriminant will always remain positive. From this, we expect that the model will always behave monotonically.

To check when the eigenvalues remain negative, we look for conditions when $(b+d)$ is greater than $\sqrt{(b-d)^2 + 4ac}$ in magnitude. After doing some algebra, the result turns out to be: $bd > ac$. In the context of the scenario, the arms spending of countries P and G will converge to complete mutual disarmament if their combined expenditure burden outweighs the combined mutual fear they have for each other.

On the other hand, if $bd < ac$, then the combined mutual fear will outweigh the combined expenditure burden. In this case, the equilibrium of mutual disarmament will become unstable and both nations will engage in a perpetual arms race.

Another point to consider is what happens when $bd = ac$. In this case, we substitute $ac = bd$ in the formula for the eigenvalues:

$$\lambda = \frac{-(b+d) \pm \sqrt{(b+d)^2 - 4(bd-ac)}}{2} \lambda = 0, -(b+d)$$

In this case, one eigenvalue is negative and the other is 0. The closed form sheds further light onto what this implies for the model. To this end, the eigenvalues were substitued back in the Jacobian to derive the eigenvectors.

These eigenvectors are

$$\tilde{v}_1 = \begin{bmatrix} \frac{a}{b} \\ \\ 1 \end{bmatrix}$$

for $\lambda_1 = 0$ and

$$\tilde{v}_2 = \begin{bmatrix} -\frac{a}{d} \\ \\ 1 \end{bmatrix}$$

for $\lambda_2 = -(b+d)$.

The closed form solution is therefore:

$$\begin{bmatrix} p(t) \\ g(t) \end{bmatrix} = \begin{bmatrix} \frac{a}{b} & -\frac{a}{d} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e^{(0)t} & 0 \\ 0 & e^{-(b+d)t} \end{bmatrix} \begin{bmatrix} \frac{a}{b} & -\frac{a}{d} \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} p(0) \\ g(0) \end{bmatrix}$$

To determine the long-term behavior of the closed form, we let $t \to \infty$, which results in the following convergence.

$$\begin{bmatrix} p(\infty) \\ g(\infty) \end{bmatrix} = \begin{bmatrix} \frac{dp(0)+ag(0)}{b+d} \\ \frac{cp(0)+bg(0)}{b+d} \end{bmatrix}$$

Within the scope of the scenario, this convergence suggests that both countries will reach a stabilized level of spending contingent on what they were initially spending.

To summarize the analysis, the base model without any underlying trust or hostily has three types of convergences. Attached alongside each explanation of these convergences are vector plots to illustrate the scenarios.

- When $bd > ac$, both countries mutually disarm ((0,0) is a stable equilibrium point) (Figure 1)
- When $bd < ac$, both countries engage in an a perpetual arms race ((0,0) is an unstable equilibrium point) (Figure 2)
- When $bd = ac$, both countries reach a stabilized level of spending contingent on initial spending (There is a line of stable equilibrium points) (Figure 3)

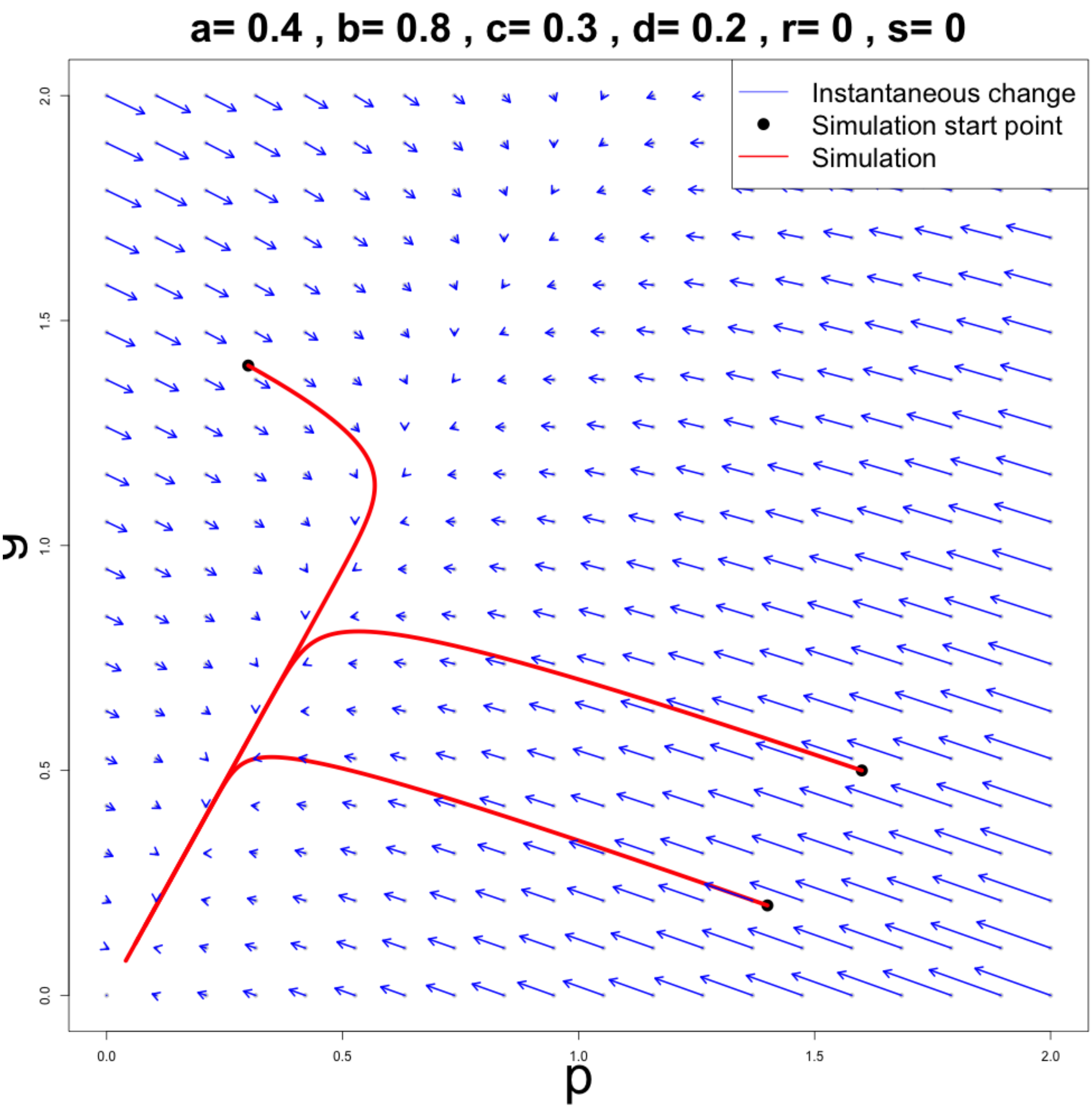

Figure 1: Mutual Disarmament

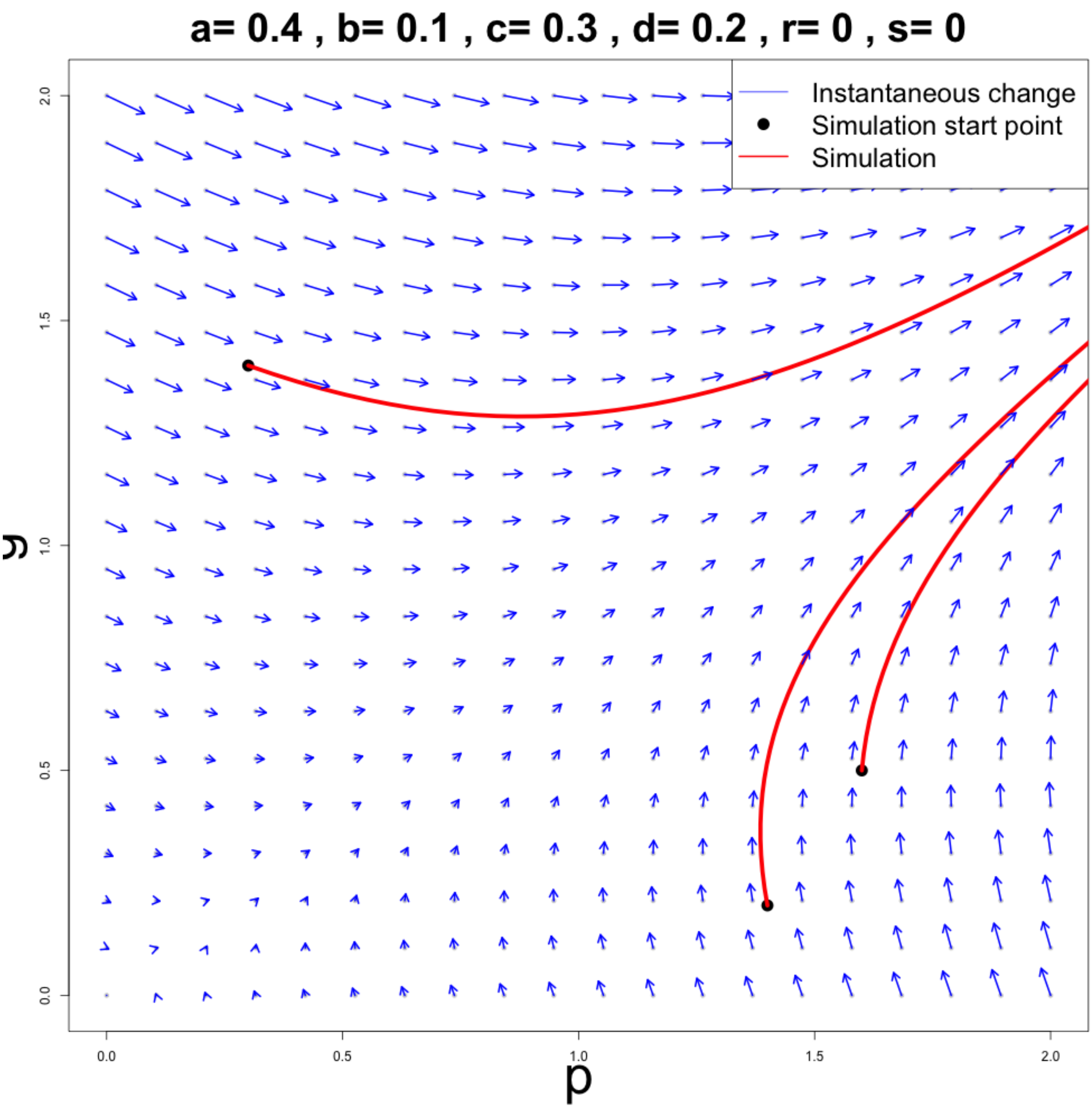

Figure 2: Perpetual Arms Race

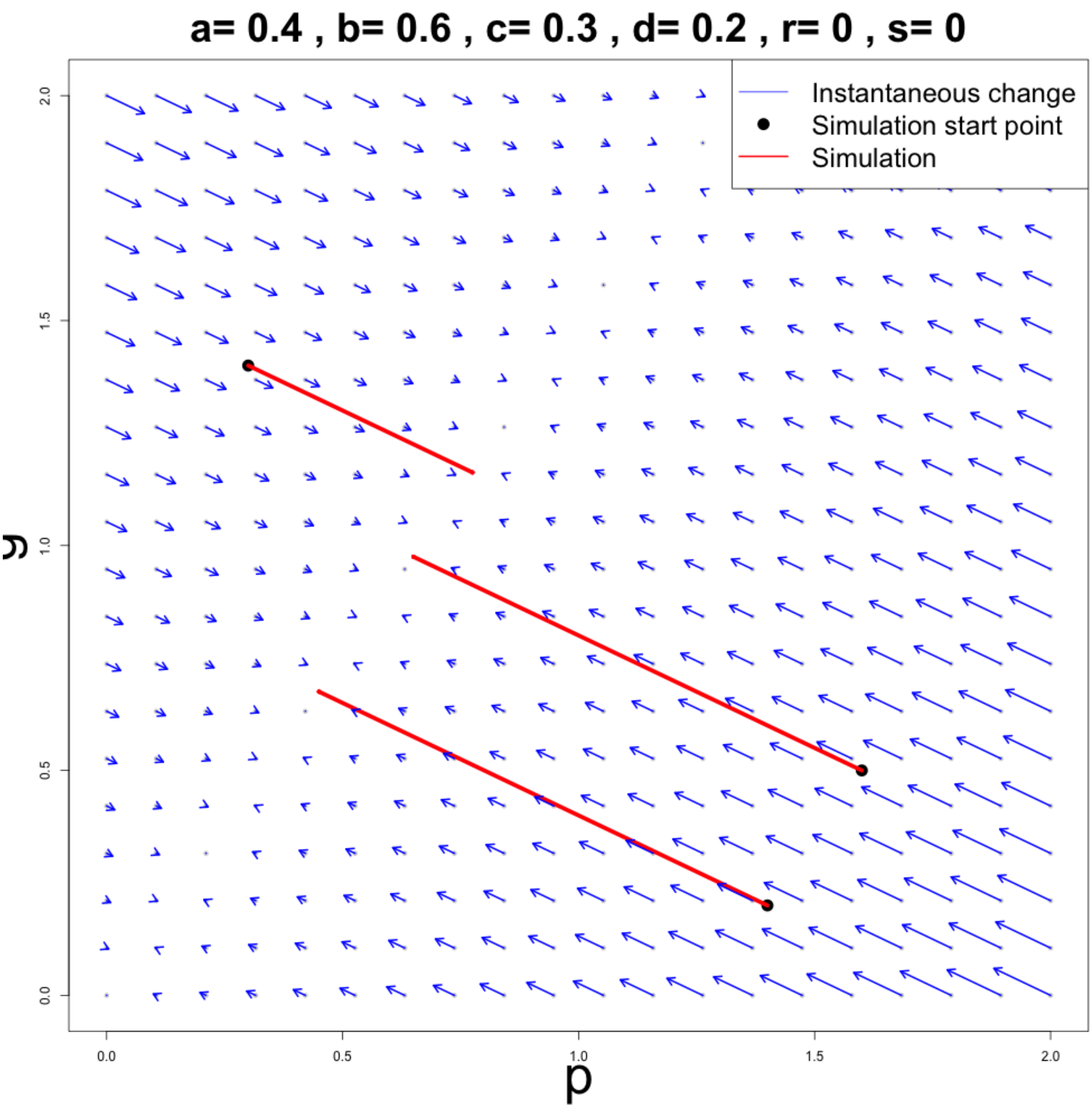

Figure 3: Stable Spending

Analysis with trust or hostility

Now that we assume r, s to be some other value other than zero, the model is not homogenous anymore, meaning that there is not a convenient closed form solution of the system as there was before.

The equilibrium point also changes:

$$\begin{bmatrix} p^*(t) \\ \\ g^*(t) \end{bmatrix} = \begin{bmatrix} \frac{rd+sa}{bd-ac} \\ \\ \frac{rc+sb}{bd-ac} \end{bmatrix}$$

Now the convergences of the model may not necessarily converge to zero. To determine the stability conditions of this equilibrium, the Jacobian from the previous form of the model can be reused as the system is still linear. In this case, we have the same stability conditions for the equilibrium point as before: $bd > ac$.

To determine what happens when $bd = ac$, we conduct simulations to adjust the values of r and s.

> This subsection should describe the various analysis you've performed on your base model and the results you've obtained from them. You should be using a mix of analytical tools (e.g., finding equilibrium points, determining stability) and numerical tools (e.g., model sim- ulations in R). When discussing analytical calculations, you don't need to list out every single step of your calculation; just present the important/key steps. When discussing nu- merical simulations, you should describe what your R code is doing and present some plots to supplement your written descriptions; all R code should be presented in an appendix at the end of the paper (see below). Be sure to describe what these analysis tell you

about your base model and what predictions they make (this should be in the context of the real-world scenario framing your project).

Model Extension

Here, you should describe the extension you've chosen to study for your model. Start by discussing what the extension is and how you plan to change your base model to account for it. Similar to the base model subsection, be sure to talk about any assumptions in your modelling, define state variables and parameters (including units), and describe any modelling choices made. In particular, you should highlight how your extended model differs from your base model. If appropriate, it would be good to include a model diagram. You should also write out the model equations. Do not discuss any analysis of the extended model here. That will go in the next section

Results

n this section, you should describe the analysis you performed on your extended model. Be sure to motivate your analysis in the context of your research question (i.e., why do you want to perform each analysis, what are you hoping it will tell you). These analysis can be purely simulation-based: you are not required to perform any mathematical analysis of your extended models, but if you can, you should! Mathematical analysis generally tell you more about model behaviour than simulations (when both are possible). You should be including plots here, as well, to go along with any simulations you did. Unlike with the draft, this section of the paper should now be complete and all results should be discussed in the pape

Discussion

This is the final section of the main body of your paper. Start by briefly summarizing the main results you presented in the last section and relating them back to your research question. This is also where you should be providing answers, based on your results, to your research question(s). Be sure to interpret your results in the context of the real-world scenario framing your project. What are the implications of your results?

You should also discuss limitations of your work. This is usually related to the modelling assumptions you made before. When are your results applicable and when are they not applicable? What factors have you omitted from your model that may affect the predictions? This is important to discuss, otherwise the reader may come away from your paper with the impression that your results are universal which is rarely the case.

Finally, you should discuss possible directions for future research. Did your results open up any other questions? Where else could you go with this work? Are there other things you've read about in the literature that could be interesting to look at? You should also use this as a way to conclude your whole paper.

Appendix: Individual Contributions

Zeyn Jaswal

1. Presentation: Overview
2. Report: Introduction and Conclusion

Azzaam Khan

1. Presentation: Base Model
2. Report: Base Model

Timothy Palin

1. Presentation: Results
2. Report: Results

Tony Xu

1. Presentation: Logistic Mutual Fear Extension
2. Report: Logistic Mutual Fear Extension

Meredith Reeves

1. Presentation: Public Sentiment
2. Report: Public Sentiment

References